

CYBERKINKS

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1. Explore/Watch the attached supplemental video in the uploaded PDF lecture notes for week 13-14.

2. Then answer the following questions below:

A. What are the two (2) types of fragmentation? Differentiate the two.

- **Internal Fragmentation**

- It happens when the memory given to a program is a bit more than what the program needs. The extra unused space is wasted.
- Even though a small amount of wasted space might not seem like much, it adds up when many programs waste memory, it can lead to significant inefficiency in memory utilization.

- **External Fragmentation**

- This happens when small free memory blocks are scattered around, making it hard to find big enough spaces for new programs. This fragmentation can make it difficult to efficiently use the available memory.
- It can lead to inefficient memory usage and can make it difficult to allocate large blocks of memory to processes.

B. What are the differences of virtual and physical memory?

- **Virtual Memory**

- An imaginary memory space created by the computer.
- Acts like extra RAM by using part of the hard drive.
- Also, it allows programs to run even if you don't have enough actual RAM.
- They are potentially much larger than physical memory, limited by the available disk space.

- **Physical Memory**

- It is an actual RAM chips inside a computer.
- It stores data and programs currently being used.
- Provides fast access to data, making your computer run smoothly.
- They are limited by the physical number of memory modules installed

C. Differentiate page and frame.

- **Page**

- They are logical units of memory used by the operating system and applications
- It is a fixed-size block of virtual memory, typically ranging from 4KB to 64KB.
- It exists in the virtual address space, which is a conceptual address space that a process sees.

- Each application program has its own view of the memory, known as logical memory.
- The operating system's memory manager handles them. When not in use, they are kept on a secondary storage device like a hard disk.
- **Frame**
 - They are physical units of memory used by the computer's hardware
 - It is a fixed-size part of physical memory, usually the same size as a page. It's a way to divide the main memory.
 - It exists in the physical address space, which is the actual memory hardware.
 - They are managed by the computer's memory management hardware and are directly accessed by the CPU.

D. Discuss the process of paging.

- First, it divides the memory. The computer's memory is split into fixed-size units called "pages" for the program and "frames" for physical memory. Then, it maps pages and frames, where each page of the program is stored in a frame in physical memory. The operating system keeps track of where each page is stored. After that, operating system uses a page table to remember the location of each page. This helps in translating the program's page numbers to the physical memory frame numbers. Lastly, when the program needs to access data, it uses the page table to find out where the data is in physical memory. If the needed page isn't in physical memory, the operating system retrieves it from secondary storage and loads it into a free frame.